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Two Predator–Prey Difference Equations Considering Delayed Population Growth and Starvation†

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A predator-prey model with overlapping generations is proposed, assuming delayed population growth and a new estimate of the number of predators starving to death. Differences among simulation results caused by manipulation of values of coefficients are discussed. The main results are: (i) although conceptually predator density is not associated so closely with prey density as in several other models (mainly by mortality of predators and not by their natality), the model shows only a few changes in its dynamic properties. (ii) Both the capacity of predators to resist starvation and the carrying capacity of prey have an important influence on the stability of the simulation results. (iii) Stable limit cycles were found without assuming influences of carrying capacity.

1. Introduction

The usual discrete deterministic predator–prey model (in short *ppm*) representing an analogue of the differential equations of Lotka (1925) and Volterra (1931), e.g. Hadeler (1974), Maynard Smith (1974) is described by the following equations:

$$\begin{aligned} x_{n+1} &= (1+a)x_n - bx_n y_n - ex_n^2, \\ y_{n+1} &= cx_n y_n + (1-d)y_n, \end{aligned} \tag{1}$$

with a, b, $c > 0$; $e \geqslant 0$ and $0 \leqslant d \leqslant 1$. x_n represents the prey, y_n the predator density at time n, a the prey's rate of increase and e its coefficient of capacity leading to its carrying capacity a/e, if $e \neq 0$. Each predator's capture rate is represented by bx_n which offers it the chance to have cx_n offspring, d is the predator's mortality. Without loss of generality the length of time is set at 1. Then d $\neq$ 1 for overlapping generations and $d = 1$ for non-overlapping generations. Equation (1) represents a simplification of a

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predator–prey relationship, as already pointed out by Lotka (1925) and Volterra (1931) and is based on the following assumptions:

(i) In the absence of predation, prey reproduce according to the Pearl–Verhulst equation, leading to exponential ($e = 0$) or logistic growth ($e \neq 0$). This has led, at least in several species, to a rather good fit between observations and prognosis, e.g. Allee, Emerson, Park, Park & Schmidt (1949), Gause (1934).

(ii) The average capture rate per predator depends linearly upon prey density. However, the functional response of predators to their prey at different densities was the main object of many studies leading to several other dependencies (Ivlev, 1961; Holling, 1965; Rosenzweig, 1971; Hassell & Varley, 1969). This influence on the dynamics of *ppm* has been discussed in several papers, e.g. May (1973). In order to examine these effects in the model proposed here, three types of capture rates will be considered (Fig. 1).

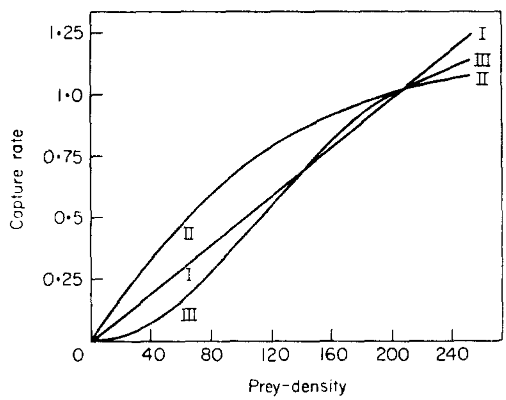

FIG. 1. Three types of average capture rates per predator and unit of time depending on the prey density. For simulations the following capture rates were used: Type I $\varphi(x) = \beta x$, Type II $\varphi(x) = \beta[1-\exp(-0{\cdot}00823x)]$ and Type III $\varphi(x)=\beta[1-\exp(-0{\cdot}0004x^2)]$.

(iii) The rate of increase of the predator population is directly proportional to both predator and prey densities. By contrast, this description of increase has been modified in several papers (Leslie, 1948; Leslie & Gower, 1960; Pielou, 1969; May, 1973). Also in the model proposed here predator reproduction can either be the same as that of the prey [see (i)] or it can depend non-linearly on both predator and prey density, which is more likely.

(iv) The rate of predator mortality is independent of prey density. Strictly speaking this seems to be justified only if a predator species has a stable age distribution and prey density oscillates only slightly. There is no doubt that a mortality rate independent of prey density permits us to assess sufficiently accurately the number of predators that die of old age, but by doing this the number of those that starved to death is fixed or ignored (Royama, 1971). By contrast, the model proposed here considers also the latter mortality factor which may be of considerable importance.

These four assumptions are commonly justified by being the first (and second) Taylor approximation(s) of really existing functional dependencies. This does not invalidate the objection under point (iv).

(v) All the influences in equation (1) appear with same time-lag. In reality both the number of offspring and the number of dying animals within a population depend on past factors, e.g. maturation of gametes, embryo development, etc., and the past history of prey density. Volterra (1931) had studied differential equations with continuous time-lags and later Brelot (1931), Nicholson & Bailey (1935), Wangersky & Cunningham (1957), Caswell (1972), Rescigno & Richardson (1973), May (1973). These past influences will also be represented in the following model.

The aim of this paper is to examine whether and how the different assumptions from equation (1) can alter the dynamic behaviour of a *ppm*.

2. The Model

To simplify the following equations we shall represent z_{n+1} as the increase in the number of predators in the time interval $\leqslant n, n+1>$ with:

$$z_{n+1} = (gy_{n-j} - hy_{n-j}^2)\{1 - \exp[-r\varphi^2(x_{n-j})]\}. \tag{2}$$

$g > 0$ is the predator's rate of increase (not net increase) and $h \geqslant 0$ its coefficient of capacity [see point (iii)] appearing with a time-lag of the length $j \geqslant 0$, which can be arranged as an integer by variation of the time units. This would lead to retarded exponential or logistic growth, if the capture rate were to remain constant. But if the actual capture rate $\varphi(x) \geqslant 0$ were low during the time of egg deposition $n-j$, few descendants per predator at time $n+1$ could be expected (y remains constant). Above a certain prey density (which is determined by $r > 0$) the real rate of increase would remain constant. This dependency could be achieved by consideration of the second factor in equation (2). Naturally this implies a simplification, since usually resources accumulated earlier and later than $n-j$ are of great importance for reproduction. It should be pointed out that this dependency between predator increase and prey density becomes weaker with increasing density of prey. Therefore, and because of time-lag, it is less than in equation (1). This can have an important influence on the stability of the proposed *ppm*. Now the *ppm* with overlapping generations can be written as:

$$\begin{aligned} x_{n+1} &= ax_{n-i} - ex_{n-i}^2 - \varphi(x_n)y_n + (1-f)x_n, \\ y_{n+1} &= z_{n+1} + (1-d)y_n - \exp\left[\sum_{k=0}^{m-1} \varphi(x_{n-k})\right](1-d)^m \times \\ &\quad \times [y_{n-m}(1-d)\{1-\exp[-\varphi(x_{n-m})\}+z_{n-m+1}], \end{aligned} \tag{3}$$

with $0 \leqslant f, d < 1$ and integers $i \geqslant 0$ and $m > 0$. (As for j these restrictions are no loss of generality.) Significance and restrictions of symbols earlier mentioned remain unchanged. f is for the sake of simplicity a constant mortality rate of the prey, i the time delay of the growth of prey and m the maximum capacity of predators to resist starvation. Predator density decreases because of death due to old age. However, the proposed model also considers death caused by starvation. An individual predator that does not die of old age in time interval $\leqslant n, n+1 >$, will in the model, starve to death during this time, if it does not capture any prey in a preceding period whose duration is greater than the predator's average capacity to resist starvation. For estimating the density of these predators, the entire predator population is divided into "starvation classes", instead of age classes (Leslie, 1945). The starvation class y_n^k consists of those predators at time n, whose starvation time is lying between k and $k+1$. Therefore $0 \leqslant k \leqslant m-1$. With equation (2) we have:

$$y_{n+1} = z_{n+1} + \sum_{k=0}^{m-1} y_{n+1}^k. \tag{4}$$

Changes in densities within the starvation classes can be written in matrix representation:

$$\begin{pmatrix} y_{n+1}^0 \\ y_{n+1}^1 \\ y_{n+1}^2 \\ \cdot \\ \cdot \\ \cdot \\ y_{n+1}^{m-1} \end{pmatrix} = (1-d) \begin{pmatrix} 1-p_n & 1-p_n & \cdots & 1-p_n & 1-p_n \\ p_n & 0 & \cdots & 0 & 0 \\ 0 & p_n & \cdots & 0 & 0 \\ \cdot & \cdot & & \cdot & \cdot \\ \cdot & \cdot & & \cdot & \cdot \\ \cdot & \cdot & & \cdot & \cdot \\ 0 & 0 & \cdots & p_n & 0 \end{pmatrix} \begin{pmatrix} y_n^0 \\ y_n^1 \\ y_n^2 \\ \cdot \\ \cdot \\ \cdot \\ y_n^{m-1} \end{pmatrix} + \begin{pmatrix} z_{n+1} \\ 0 \\ 0 \\ \cdot \\ \cdot \\ \cdot \\ 0 \end{pmatrix} \tag{5}$$

where $p_n = \exp[-\varphi(x_n)]$ is the probability of starving in interval $\leqslant n, n+1 >$. It is easy to verify:

$$y_{n+1} = z_{n+1} + (1-d)(y_n - p_n y_n^{m-1}).$$

Since:

$$y_n^{m-1} = (1-d)^{m-1} y_{n-m+1}^0 \prod_{k=0}^{m-1} p_{n-k}$$

and

$$y_{n-m+1}^0 = (1-d)y_{n-m}(1-p_{n-m}) + z_{n-m+1},$$

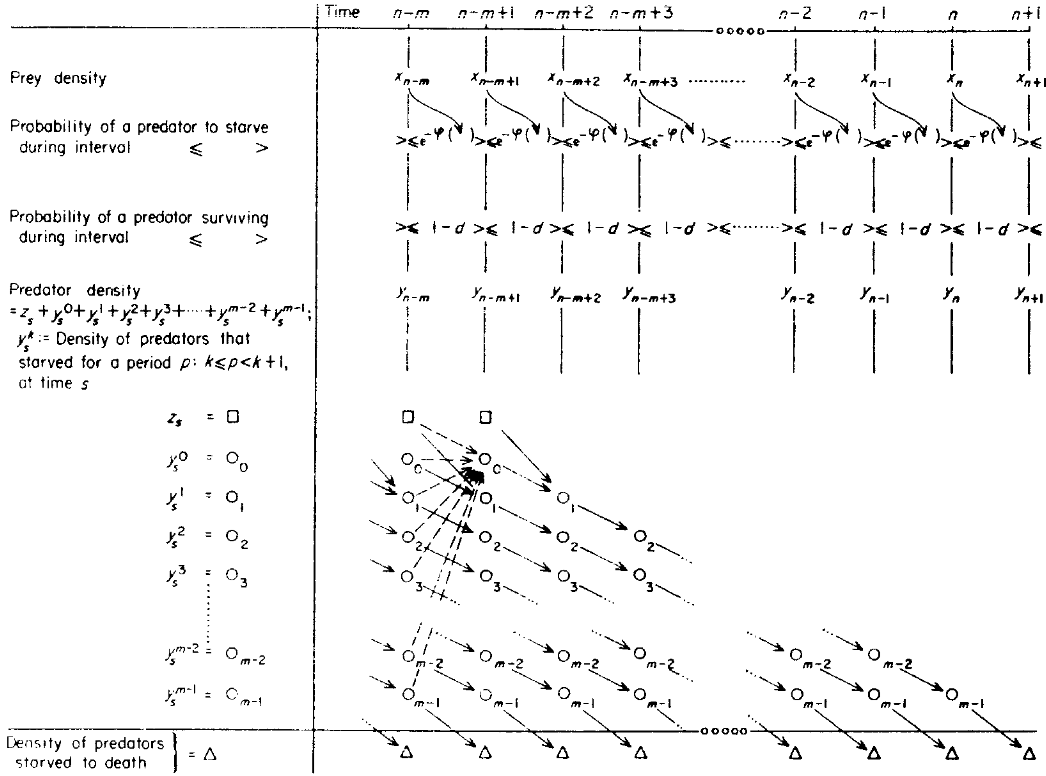

FIG. 2. Deduction of the number of predators that starved to death in interval $\leq n, n+1>$. For each point of time, the prey density x and the entire predator density y, consisting of newborn predators □ and the adult predators in the starvation classes $\bigcirc_0, \bigcirc_1, \ldots$, are specified. Solid lines indicate the probability not to die of old age, but to starve during this interval. These predators become members of the following starvation class which occurs with the probability $(1-d)\exp[-\varphi(x..)]$. Dashed lines (for a better lucidity only plotted in interval $\leq n-m, n-m+1>$) indicate the possibility not to die of old age, but to hunt successfully. In this way a predator will become member of y^0_{n-m+1} with the probability of $(1-d)\{1-\exp[-\varphi(x_{n-m})]\}$. If a predator starves to death exactly at time $n+1$ (thus it shall be member of △ to the right downwards), it must be member of $\bigcirc_0$ or □ at time $n-m+1$ and then run through the diagonal.

the description of the total predator density is identical in equation (5) and (3). Therefore, a predator can starve to death in time interval $\leqslant n, n+1>$, if it is (a) born at time $n-m+1$ and then does not die of old age but starves until time $n+1$, or, it is (b) adult and captures prey at time $n-m$ and does not die of old age but starves until time $n+1$ (Fig. 2). This calculation is based on two assumptions:

(1) The Poisson distribution yields the probability that some predators will starve because of differences in individual hunting success. This seems to be justified because of the weak convergence of the binomial distribution $B(n, \varphi(x)/n)$ against the Poisson distribution $P[\varphi(x)]$ (Krickeberg, 1963).

(2) The probability of starving is equal for members of all starvation classes. This condition is met only in a few cases, e.g. for sit-and-wait predators; for other predators the starvation time could enhance the motivation for hunting, or the accompanying physical weakening could lower the hunting success (such dependencies could easily be arranged by different weighting of the columns in the matrix).

It should be noted that equation (3) can be transformed into differential equations by adequate limit considerations.

3. Dynamic Behaviour of the Model

In many *ppm* described by differential equations it is possible to apply a theorem of Kolmogoroff (1931) (see also Scudo, 1971; Rescigno & Richardson, 1973; May, 1973) that provides sufficient conditions for either stability of the equilibrium point or the existence of limit cycles. But, because of time delays this theorem is not applicable. In addition to this handicap, an equilibrium analysis of equation (3) shows that the equilibrium point $(\hat{x}, \hat{y})$ with $\hat{y} > 0$ could be calculated only numerically. Therefore, a conventional neighbourhood analysis was omitted. Instead, many simulations were made with a Telefunken digital computer using equation (3) to investigate its dynamic behaviour. A fixed set of values of coefficients are called standard conditions and only few of them were replaced simultaneously. This was done to classify emerging equilibrium points $(\hat{x}, \hat{y})$ with $\hat{y} > 0$ and to determine possible limit cycles. If at least one of the densities was removed from $(\hat{x}, \hat{y})$, the result would be:

* Both densities return oscillating; $(\hat{x}, \hat{y})$ is a stable focus, e.g. Fig. 3, Type III in the upper row.
+ Both densities return directly (after having smoothed starting conditions); $(\hat{x}, \hat{y})$ is a stable node (not illustrated).

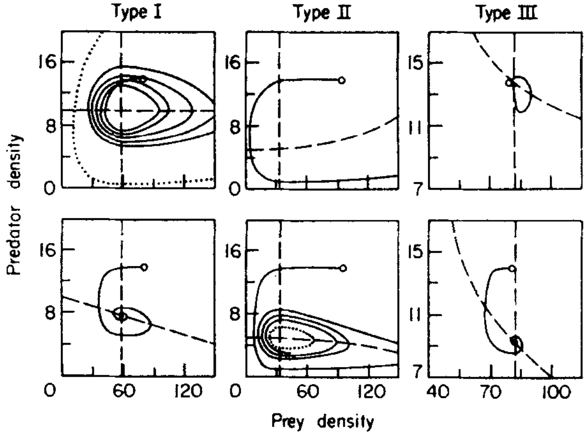

FIG. 3. Dynamic behaviour of model (3) dependent on the type of capture rate and prey capacity. In each cell the corresponding isoclines are drawn with dashed lines and potential limit cycles with dotted lines. The solid lines show the state of the system as a function of time. ○ indicates the starting point. Assumptions: $a=0{\cdot}05$, $f=i=0$, $g=0{\cdot}01$, $d=h=0$, $r=\infty$, $m=11$, $j=26$; $\beta=0{\cdot}005$ to the left, otherwise $\beta=1{\cdot}25$. In the upper row $e=0$, in the lower row $e=0{\cdot}0002$.

– Both densities oscillate around $\hat{x}$ and $\hat{y}$ with steadily increasing amplitudes and there is no periodic solution. $(\hat{x}, \hat{y})$ is an unstable focus. (0, 0) is a saddle point; both species die out asymptotically (not illustrated).

/ Both densities oscillate around $\hat{x}$ and $\hat{y}$ in the beginning, but they steadily increase after their lowest level, e.g. Fig. 3, Type II in the upper row.

○ Both densities oscillate around $\hat{x}$ and $\hat{y}$, converging against stable periodic solutions or so-called limit cycles, e.g. Fig. 3, Type II in the lower row.

The results of the simulations are characterized in Tables 1–3 employing the above symbols. In all cases characterized by "–" the result was a steady increase in amplitude of the oscillations of both predator and prey densities. Therefore no periodic solution was obtained. Predator densities fell below 0·01 in these cases. An estimation of the proportion of minimal to mean predator densities in limit cycles, as found for *ppm* described by differential equations without time delays, (see May, 1973; De Angelis, 1975) is still lacking. In the simulations the length of the periods varies almost between 200 and 300 days (= units of time). On the grounds of Tables 1–3 and the knowledge of the exact courses of the simulations, it may be concluded that:

TABLES 1–3

Behaviour of model (3) *near the equilibrium point* ($\hat{x}$, $\hat{y}$) *with* $\hat{y} > 0$. *Symbols are explained in the text. Standard conditions are:* a = 0·05, i = 0, f = 0, $\varphi = \varphi_I$ *with* β = 0·005, g = 0·01, j = 26, d = 0, m = 11 *and* r = ∞. *Deviations from these conditions are given in the Tables.*

TABLE 1

No.		I	II	III	IV
	a/e=	$a/0$	250	$a/0$	250
	g/e=	$g/0$	$g/0$	25	25
1	-S.C.-	○	*	*	*
2	a=0·1	○	*	*	*
3	a=0·025	○	*	*	*
4	i=13	*	*	*	*
5	i=20	*	*	*	*
6	i=40	*	*	*	*
7	f=0·02	○	*	*	*
8	f=0·02, i=13	*	*	*	*
9	φ_{II}, β=1·25	/	○	/	*
10	φ_{III}, β=1·25	*	*	*	*
11	φ_I, β=0·01	○	*	○	*
12	φ_I, β=0·0075	○	*	○	*
13	φ_I, β=0·0025	○	*	*	*
14	φ_I, β=0·001	○			
15	g=0·025	*	*	*	*
16	g=0·005	—	*	○	*
17	j=18	—	*	○	*
18	j=34	○	*	*	*
19	d=0·0025	○	*	*	*
20	m=7	○	*	*	*
21	m=15	—	*	○	*
22	r=5	—	*	○	*
23	r=10	—	*	○	*
24	r=15	—	*	○	*
25	r=50	—	*	○	*

TABLE 2

No.		$h=0$	$h=0{\cdot}0004$
1	e=0·0005 (a/e=100)	*	*
2	e=0·0002 (a/e=250)	*	*
3	e=0·00005(a/e=1000)	*	*
4	e=0·00001(a/e=5000)	○	*
5	e=0 (a/e=a/0)	○	*

TABLE 3

No.		e=0	e=0·0002
1	$h=0{\cdot}001$ ($g/h=10$)	+	*
2	$h=0{\cdot}0004$ ($g/h=25$)	*	*
3	$h=0{\cdot}0002$ ($g/h=50$)	○	*
4	$h=0{\cdot}00005$($g/h=200$)	○	*
5	$h=0$ ($g/h=g/0$)	○	*

(i) Stable limit cycles were attained mostly in the absence of self limitation.

(ii) The higher the prey's carrying capacity, the greater will be both the amplitudes and the duration of the oscillations of predator and prey. When exceeding a critical value, the stable focus ($\hat{x}$, $\hat{y}$) becomes unstable with the occurrence of limit cycles (Table 2).

(iii) In analogy to point (ii): The higher the predator's carrying capacity, the more unstable become the simulation results (Table 3), as shown by periodic rather than stable solutions and an increase in the amplitude of oscillations.

(iv) The time-lag i induces noticeable changes only if e $=$ 0 (exponential growth of the prey). If i is small, results will be less stable than those with increased i (Table 1, Nos. 1, 4, 5 and 6), shown by an increase in the rate of convergence. If there are no predators, the prey's carrying capacity is a stable equilibrium point at corresponding values of i. One has to suppose that a large increase in i above this value will have a destabilizing influence.

(v) The type of the functional response (at comparable values) has a great influence on the dynamics of equation (3). For a better comparison the simulation results are plotted without (Fig. 3) and with (Fig. 4) influences of carrying capacity for the predators. Predator populations hunting according to Type III induce most stable results near the equilibrium point, followed by Type I (Figs 3 and 4). A capture rate of Type II will induce a return to

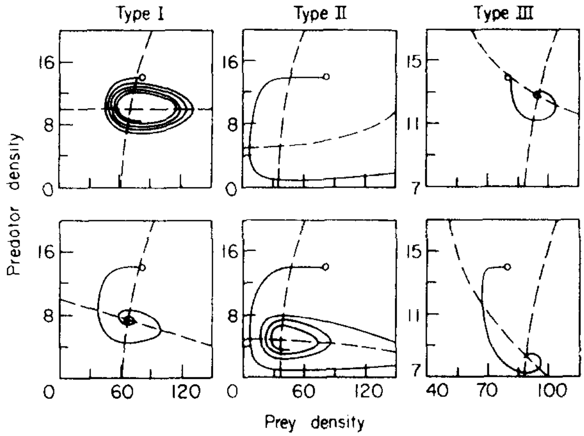

FIG. 4. Dynamic behaviour of model (3) dependent on the type of the capture rate and prey capacity. Conditions as in Fig. 3, but $h=0{\cdot}0004$.

equilibrium only if both species are limited by other resources (see also Table 1, Nos. 1, 9 and 10).

(vi) Deviations in efficiency of hunting from the standard conditions always produce destabilizing effects with increasing capture rate. This holds for both directions (Table 1, Nos. 11, 12, 1, 13 and 14). In Table 1, No. 14, II–IV, equilibrium points are lacking. Simulations show steadily increasing prey and constant predator densities at its carrying capacity for II, while in III and IV the prey reaches its carrying capacity and the predator dies out.

(vii) It is interesting that a reduction of the time delay for the predator j has a destabilizing influence compared with those results under standard conditions, whereas an extension of j affects stabilization, shown by smaller amplitudes in the oscillations (Table 1, Nos. 17 and 18).

(viii) As expected, the capacity to resist starvation affects the degree of linkage between prey and predator densities. The smaller m the tighter the linkage; the larger m the sooner the prey is over-exploited, which leads to greater oscillations (Fig. 5). The results of Table 1, Nos. 20 and 21 show the same tendency.

(ix) The proposed dependency of the rate of predator increase on prey density [see equation (2)] causes an unexpected destabilization as compared with those cases, in which it is lacking (Table 1, Nos. 22–25). It was thought that a reduction of newborn predator individuals during low prey densities would have a stabilizing influence, but the opposite occurs. Prey density

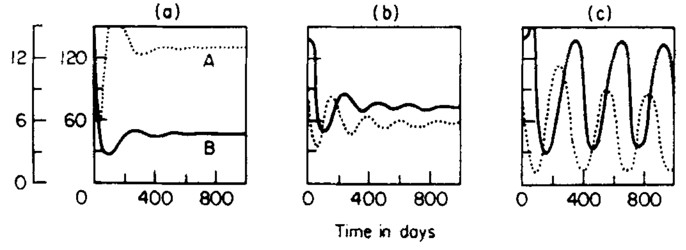

FIG. 5. Dynamic behaviour of model (3) for alternative starvation capacities of the predators. Values of coefficients as in Fig. 3 to the right and beneath, but (a) $m=6$, (b) $m=11$ and (c) $m=21$. A denotes the prey—, B the predator density.

did not sink to such a low level but this offers the chance for a larger increase of prey density, strengthened by the time delay. This process can be slowed down only after a longer time accompanied by higher predator densities. Thus a self-stimulating process commences.

4. Conclusions

Equation (3) is conceived as a *ppm* for overlapping generations and therefore differs from *ppm* of Maynard Smith & Slatkin (1973), Beddington, Free & Lawton (1975), since an influence of starvation between generations is difficult to imagine. In order to describe overlapping generations, one must change the magnitude of the parameters, so that equation (3) is shifted towards a pair of differential equations. This is expressed in its dynamic behaviour too (differential equations which are analogous to equation (3) would not be as easy to handle numerically and do not offer more valuable information). In comparing equation (3) with other differential equations describing *ppm* one finds an essential difference. In most of them only the number of newborn predators and not the number of dying animals depends on prey density.

Differential equations with time delay for predators and prey are those of Volterra (1931), who considered no carrying capacities and the equations of Brelot (1931), who did. In the former case, existing equilibrium points are always unstable and it is not possible to prove that non-stationary solutions are periodic. Under corresponding assumptions for equation (3) there are both stable and unstable equilibrium points, the latter of which may or may not have periodic solutions. In the latter case, existing equilibrium points will not necessarily be stable, but the solution will be bounded, if there are influences from carrying capacity of the prey. This agrees with results obtained from equation (3) [see point (i)].

In essence, equation (3) resembles best a model of Leslie (1948), an extension of which has been examined by Leslie & Gower (1960) and Pielou (1969). Although time delays are lacking, the decrease and not the increase of predators depends on the prey density. For these models stable equilibrium points are always found. May (1973) examined this model with a slight modification of the capture rate and found that Kolmogoroff's theorem is applicable. Therefore, either non-trivial equilibrium points are stable or there are periodic solutions. Caswell (1972) modified the model of Leslie (1948) incorporating time-lags and a capture rate defined by a measure of hunger. His simulations show that for a stable co-existence of prey and predator refuges for the prey are necessary. In equation (3) this additional assumption can be omitted.

Results (ii) and (iii) are found in other *ppm* too (Rosenzweig, 1971; May, 1973), well known under the name: Paradox of enrichment. It is interesting [see point (v)] that predator populations hunting by search image (which induces the sigmoid capture rate of Type III) produce more stable results than those with linear ones. Normally for a predator a capture rate of Type III should only be assumed if there were additional prey in its diet. However, such a situation could also be described by equation (3), if the density of this additional prey were to remain constant and the capacity to resist starvation were to be interpreted as a capacity to do without some essential substance provided only by the basic prey whose density is described by equation (3). In such a special case this result could be applied. On the other hand, it is well known that a capture rate of Type II tends to destabilize a predator–prey system (May, 1973). Result (iv) seems to contradict the results of Barclay & van den Driesche (1975). They made simulations of food webs at several trophic levels considering time delays and found that simultaneously existing time-lags in predator and prey have destabilizing effects no matter whether time-lags are discrete or continuous. However, they also concluded [compare point (vi)] that too-efficient predators could induce enough instability to destroy the system.

The concept underlying equation (3) is thought to be realistic at least for some predator–prey systems. In quality, equation (3) shows plausible results that are consistent with those of other models for the most part. The influence of starvation alters the dynamics in an expected manner. Both in the real world and in the model this will be noticeable if the average capture rate per unit time is not too high.

The type of dependence of the predator increase on prey density seems to have an important influence on the dynamics of *ppm*. The model needs testing in the laboratory and in the field to determine whether it is sufficiently realistic to produce reliable prognosis.

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